CTA200 Assignment #3

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Introduction. This assignment analyzes two examples of nonlinear dynamical behaviour: (*i*) the Mandelbrot set is computed by fixing points c in the complex plane, setting $z_0 = 0$, and iterating $z_{i+1} = z_i^2 + c$ to classify each point as bounded or divergent; and (*ii*) the Lorenz equations are numerically integrated to reproduce the aperiodic trajectories presented by Lorenz [1] and to demonstrate sensitivity to initial conditions. These two examples illustrate how deterministic systems can generate unpredictable long-term behaviour.

I. Question #1

The Mandelbrot set was computed by choosing points $c = x + iy$ on the complex plane over the region $-2 < x < 2$ and $-2 < y < 2$, setting $z_0 = 0$, and iterating $z_{i+1} = z_i^2 + c$. The region was sampled using a 1000×1000 grid, corresponding to 10^6 complex points, with each point iterated at most 100 times. A point was classified as divergent once $|z| = \sqrt{\Re(z)^2 + \Im(z)^2} > 2$. From this setup, two arrays are outputted: (*i*) the Boolean array `bounded`, whose entries are `True` for points that do not diverge within 100 iterations and `False` for points that do, and (*ii*) the integer array `divergence_iter`, whose entries record the iteration number at which each point diverges, with entries left as 0 for points that do not diverge. Plotting `bounded` produces a two-colour image, with bounded points shown in yellow and divergent points shown in dark purple; see Figure 1. Plotting `divergence_iter` instead assigns each point a colour according to the iteration number at which it diverges (0 for points that do not diverge); see Figure 2.

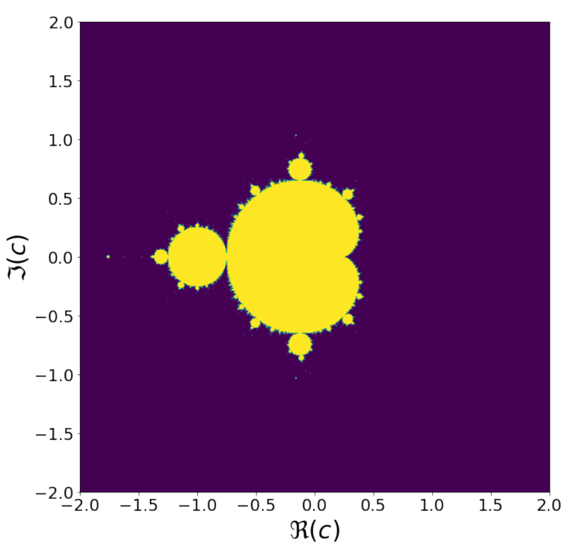

FIG. 1. Mandelbrot set computed on the complex plane for $-2 \leq \Re(c) \leq 2$ and $-2 \leq \Im(c) \leq 2$. All points are evolved under $z_{i+1} = z_i^2 + c$ for up to 100 iterations: yellow points remain bounded, dark purple points diverge.

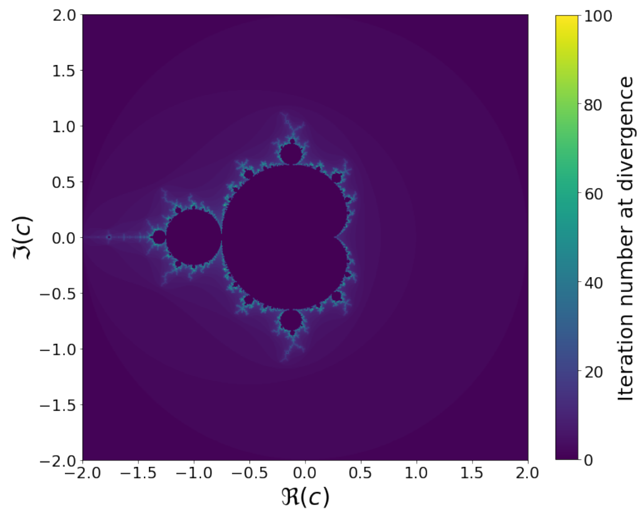

FIG. 2. Mandelbrot set coloured by the iteration number at which each point diverges under $z_{i+1} = z_i^2 + c$. Very dark regions ($= 0$) correspond to points that remain bounded through 100 iterations, dark regions ($\gtrsim 0$) correspond to rapid divergence, and brighter regions (> 0) indicate slower divergence near the boundary of the set.

II. Question #2

The Lorenz system consists of three coupled ordinary differential equations given by

$$\dot{X} = -\sigma(X - Y), \tag{1}$$

$$\dot{Y} = rX - Y - XZ, \tag{2}$$

$$\dot{Z} = -bZ + XY \tag{3}$$

where $\sigma = 10$, $r = 28$, and $b = 8/3$ are the standard parameter values used by Lorenz [1]. Here, σ is the dimensionless Prandtl number, representing the ratio of kinematic viscosity to thermal diffusivity; r is the dimensionless Rayleigh number, which depends on the vertical temperature difference across the fluid layer; and b is a dimensionless length scale. The state vector is $W = [X, Y, Z]$, with initial conditions $W_0 = [X_0, Y_0, Z_0] = [0, 1, 0]$. The system in Equation 1 was numerically integrated from $t = 0$ to $t = 60$ using `solve_ivp` from `scipy.integrate` with time-step $\Delta t = 0.01$. The itera-

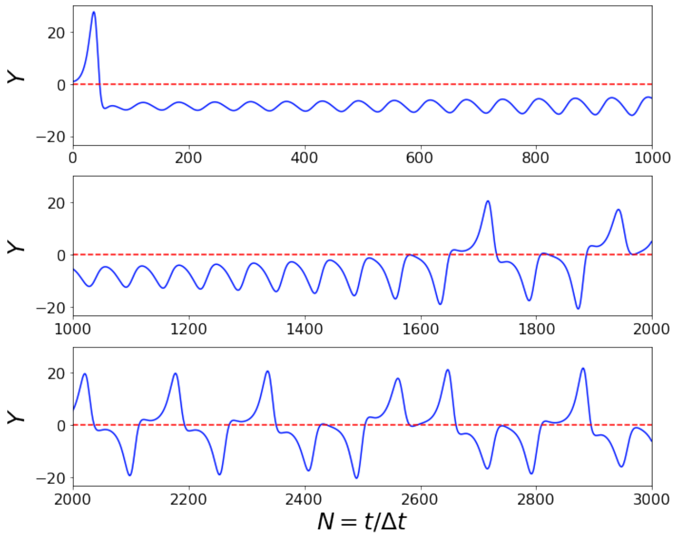

FIG. 3. Numerical solution of the Lorenz system showing Y as a function of iteration number $N = t/\Delta t$ for the first 3000 iterations; appearance is aperiodic. Solution computed using $\sigma = 10$, $r = 28$, $b = 8/3$, $W_0 = [0,\ 1,\ 0]$, and $\Delta t = 0.01$. Dashed red line marks $Y = 0$.

tion number is then given by $N = t/\Delta t$.

Figure 3 reproduces Figure 1 of Lorenz [1]. Small differences from the original figure are expected, as the Lorenz system is chaotic: tiny numerical differences can grow rapidly during evolution. Although the same initial conditions and parameter values are used here, Lorenz [1] used a different numerical integration method, hence the trajectories need not agree point-by-point at later times. Figure 4 similarly reproduces Figure 2 of Lorenz [1].

To test sensitivity to initial conditions, the Lorenz system was integrated again using the same parameter values (σ, r, b) but with the perturbed initial conditions $\tilde{W}_0 = [0,\ 1.00000001,\ 0]$; this differs from the original initial condition by 10^{-8} in the Y_0 component. The phase-space separation between the original trajectory $W(t)$ and the perturbed trajectory $\tilde{W}(t)$ at each time t is

$$d(t) = \sqrt{(\tilde{X} - X)^2 + (\tilde{Y} - Y)^2 + (\tilde{Z} - Z)^2}. \qquad (4)$$

Plotting $d(t)$ on a logarithmic scale against linear time t gives Figure 5. The approximately linear region on the semilog plot indicates approximately exponential growth of the initial perturbation, demonstrating sensitive dependence on initial conditions.

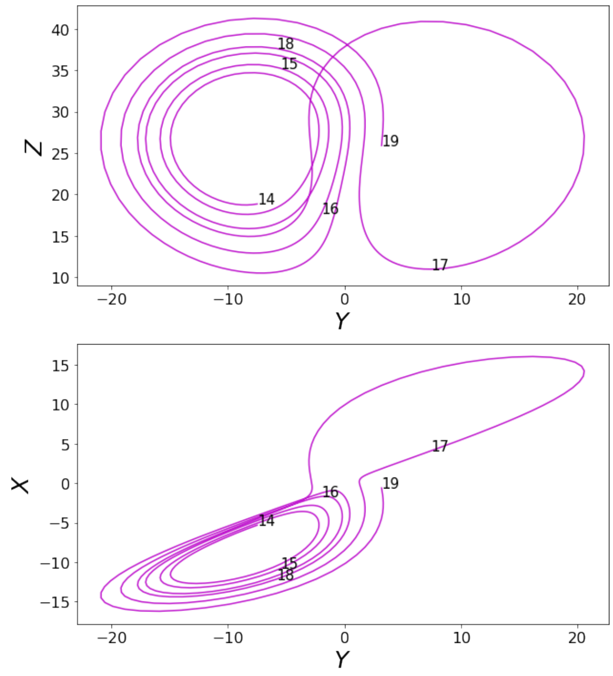

FIG. 4. Phase-space projections of the Lorenz trajectory for $14 \le t \le 19$, corresponding to $1400 \le N \le 1900$ for $\Delta t = 0.01$. Upper panel shows Y-Z projection; lower panel shows Y-X projection. Labels mark the trajectory position at times $t = 14,\ 15,\ \ldots,\ 19$.

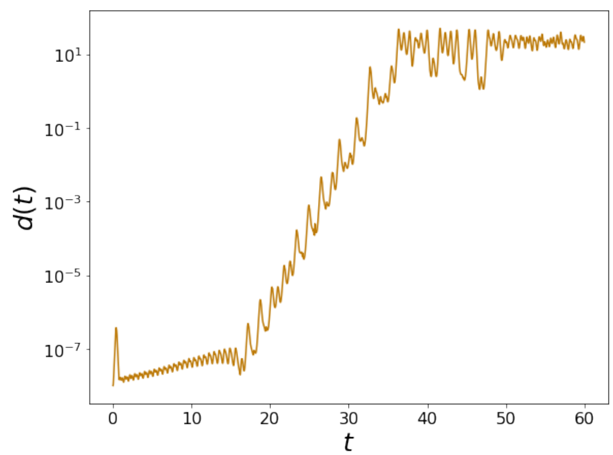

FIG. 5. Semilog plot of the phase-space separation $d(t)$ between two Lorenz trajectories with initial conditions $W_0 = [0, 1, 0]$ and $\tilde{W}_0 = [0, 1.00000001, 0]$. The approximately linear growth on the semilog scale indicates exponential amplification of the initial perturbation.

III. Conclusion

The Mandelbrot calculation shows that repeated application of $z_{i+1} = z_i^2 + c$ separates the complex plane into bounded and divergent regions with a highly structured, fractal boundary. This means that tiny changes in c near the boundary can change whether the point remains bounded or diverges, hence deterministic iteration can yield high sensitivities to initial conditions.

The Lorenz-system calculation shows that deterministic differential equations can produce aperiodic motion and unpredictable long-term behaviour. In particular, two trajectories initially separated by only 10^{-8} exhibit approximately exponential separation, providing numerical evidence of chaotic behaviour.

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[1] E. N. Lorenz, Deterministic nonperiodic flow, Journal of the Atmospheric Sciences **20**, 130 (1963).